

Scoring Signals — Computation & Penalty Detail

⚡ SIGNAL COMPUTATION — How each signal is calculated

📄 external_similarity (w=0.28)

CompositeExternalSignalProvider:
github_score = GitHubCopilotProvider.score(src, tgt)
secondary = LLM(0.60) × llm_score + KB(0.40) × kb_prior_score
composite = GitHub(0.65) × github_score + Secondary(0.35) × secondary

Falls back to 0.0 if no provider available

🔗 alias_similarity (w=0.18)

normalize(name):
strip vendor prefixes (HAL_, LL_, BSP_)
lowercase, split on _/camelCase
remove common suffixes (_IRQHandler, _Init)

score = SequenceMatcher(norm_src, norm_tgt).ratio()

🗂 context_similarity (w=0.14)

Build call-site neighbor sets:
src_neighbors = functions in same source_object
tgt_neighbors = functions in same source_object

score = |intersection| / |union| (Jaccard)

📦 object_similarity (w=0.14)

Compare source_object paths:
Split on / and \ → path tokens
Longest common subsequence of tokens
score = LCS_length / max(len_src, len_tgt)

✂ name_similarity (w=0.12)

Raw token comparison (no normalization):
src_tokens = split_camel_and_underscore(src_name)
tgt_tokens = split_camel_and_underscore(tgt_name)
score = 2 * |common_tokens| / (|src| + |tgt|) (Dice coefficient)

👤 role_similarity (w=0.08)

1.0 if src.layer == tgt.layer
0.5 if adjacent layers
0.0 otherwise

🔑 signature_similarity (w=0.06)

Size ratio: 1.0 - |src.size - tgt.size| / max(src.size, tgt.size)
Capped at 0.0 minimum

⚠ PENALTY MULTIPLIERS — Applied after weighted sum (stackable)

×0.55 — Runtime/Toolchain Pair

Trigger: Both items classified as system/runtime layer
Rationale: Runtime pairs are vendor-specific, unlikely functional match

×0.70 — Unknown-Unknown Pair

Trigger: Neither item found in any vendor registry or KB prior
Rationale: No contextual evidence, score unreliable

×0.75 — Name Match + Role Mismatch

Trigger: name_similarity > 0.8 but role_similarity == 0.0
Rationale: Same-name different-layer functions are false positives

×0.85 — Low Signal Uncertainty

Trigger: coverage < 0.4 (fewer than 3 signals contributed non-zero)
Rationale: Insufficient evidence for reliable scoring

⚠ Penalties STACK multiplicatively: if runtime (×0.55) AND unknown (×0.70) apply → effective = conf × 0.55 × 0.70 = conf × 0.385

📊 FINAL FORMULA

confidence = (Σ signal_i × weight_i) × Π penalty_j
coverage = count(signal_i > 0) / 7

Final bucket: confidence ≥ 0.72 → accepted | 0.55–0.71 → review | 0.45–0.54 → alternative | < 0.45 → rejected